AOS C111
Emilio Yanez
Dr. Alexander Lozinski
08 December 2023

Investigating the AR IVT-Precipitation Relationship Using ML Methods

Introduction

Atmospheric Rivers (ARs) are elongated and narrow filaments of water vapor transport in the atmosphere. Convergence of these water vapor plumes is often associated with heavy precipitation events (e.g., US West Coast). As part of my research in Dr. Gang Chen's group, we have quantified the relationship between an AR's moisture, wind, integrated vapor transport (IVT), and associated precipitation in two main ways. The first is by developing a global, percentile-based classification of wet and windy ARs in satellite observations (similar to [2]). The second (and the focus of this project) is using a binned scatter approach to understand better the sensitivity of mean AR precipitation to changes in other AR variables (e.g., moisture, wind, and IVT). Binned scatter plots [1] divide the independent variable into dynamically-sized bins and find the conditional means of a dependent variable, similar to the convective onset statistics for tropical convective precipitation against changes in water vapor [3]. In previous work, we found a strong conditional relationship between mean AR IVT and mean AR precipitation, regardless of the ocean basin and season. These conditional statistics in satellite observations may provide a benchmark for quantifying the relationship between ARs and precipitation in other data sets (such as reanalysis and climate models). As such, understanding and building an accurate "base" AR IVT-precipitation relationship is important for subsequent studies.

Deriving a conditional relationship between AR IVT and AR precipitation from raw data is challenging since there is considerable noise when plotting a simple scatterplot. This motivates using a machine learning (ML) approach to simplify the scatter in the data. Supervised algorithms, such as linear regression, support vector regression, and random forest regression can be used to predict mean AR precipitation from other AR variables. Once the ML model is trained, we can plot new scatter plots of mean AR IVT versus the *predicted* mean AR precipitation. The hope is that the predicted data, modeled from real satellite data, can reproduce the conditional AR IVT-precipitation relationship found in the original, non-ML method (my previous research). Overall, I find that all three ML models (linear, SVR, and random forest) predict the original conditional relationship with good accuracy.

Data

The dataset used comes from my previous work and includes six features: (0) mean AR wind, (1) mean AR integrated water vapor (IWV), (2) mean AR IVT, (–) mean AR precipitation, (3) central longitude, and (4) central latitude. Each entry in the dataset represents one AR event, with

the six variables being the average value within the AR shape. The datasets fed into the ML algorithms have already been pre-processed from raw satellite data. In short, I used code from previous research to create an AR shape mask (the geometric extent of each AR) and computed the average wind, IWV, IVT, and precipitation within each AR shape. I repeated this for each AR event in each timestep, for a complete dataset spanning from 2002–2016. The last step was to add the central coordinates of each AR and build a data frame containing all six features. Once done, I could export as .txt files and use this pre-processed data for my final project. Figure 1 shows a screenshot of what the final dataset looks like. Figure 2 below shows the raw scatter between mean AR IVT and mean AR precipitation, for November–March (NDJFM) and May–September (MJJAS) ARs in the northern hemisphere (NH).

```
[45] AR_all_NDJFM.head()
```

	AR Mean Wind (m/s)	AR Mean IWV (kg/m2)	AR Mean IVT (kg/m/s)	AR Mean Precip (mm/day)	Central Lons (deg)	Central Lats (deg)
0	11.955742	41.201134	467.200958	12.655289	264.905	21.200
1	10.812697	22.832947	229.755142	4.286365	43.350	26.029
2	21.365599	30.414259	522.013306	15.923621	136.535	30.678
3	16.168025	11.211940	157.911865	1.953020	76.008	65.506
4	19.779290	17.559341	275.033691	24.203941	338.180	59.712

Figure 1. One of the two final datasets used in my project. Contains six features: (0) mean AR wind, (1) mean AR IWV, (2) mean AR IVT, (–) mean AR precipitation, (3) central longitude, and (4) central latitude.

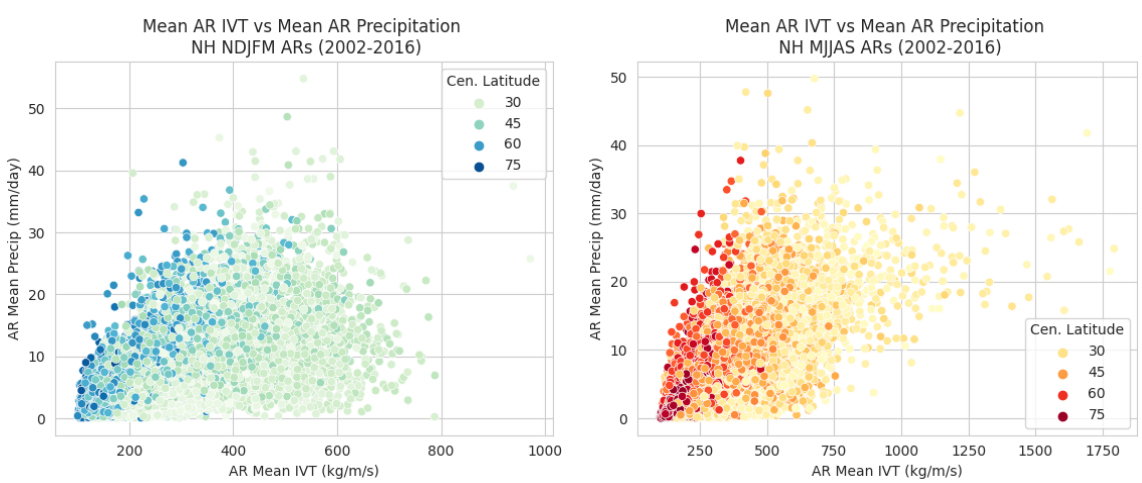

Figure 2. Mean AR IVT versus mean AR precipitation, for both NH NDJFM and NH MJJAS ARs. Notice that while we can guess there should be some positive relationship, it is hard to know just from a raw scatterplot. The color hue (central latitude in degrees) is added to highlight that generally, high latitudes correspond to lower IVT values and slightly less AR precipitation.

Machine Learning Models

The machine learning algorithms chosen were linear regression, support vector regression (SVR), and random forest regression. These are all supervised learning algorithms and are used since our goal is to predict AR precipitation from AR IVT (continuous variables). Each algorithm was implemented in Python using the `scikitlearn` library. The input datasets (X) were the original data frames without mean AR precipitation (so just five features) and the target variable (y) was the mean AR precipitation. The data was split into test (30%) and training (70%) datasets, using a fixed random state to ensure reproducibility. A standard scaler and pipelines were used to shorten the code for each algorithm. For the linear regression model, ridge regularization was used. Since I already know the underlying relationship should be roughly linear, the default parameters ($\alpha = 1$) were kept. The SVR and random forest models were used to capture any non-linear deviations from the expected linear relationship. For example, the MJJAS plot in Figure 2 shows some scatter that may be non-linear for high values of IVT (the tail end). For the SVR model, experimentation with the parameters C (regularization) and ε ($L2$ -squared penalty) showed no major differences for different parameter choices. Since the dataset is not too large, I chose $\varepsilon = 0.01$ to penalize mistakes heavily (in theory, this gives more accurate results). For the random forest model, varying the choice of maximum depth changed the output considerably. I chose a max depth of 10, which I found to be a balance between underfitting (too shallow trees) and overfitting (too deep trees). Lastly, for the models with random estimators or splitting (such as linear and random forests), I fixed the random state parameter to help keep consistent results each time I ran my code.

Once each model was trained, it was tested by outputting a predicted mean AR precipitation. The models return an estimation using all five remaining features, but for plotting purposes, only the results of mean AR IVT as a predictor of mean AR precipitation are shown. This choice is motivated by two reasons. For one, a simple covariance matrix (see code) of the full dataset shows that AR IVT is the most related to AR precipitation (it has a positive covariance value of ~0.55). Secondly, the results of the random forest model show that the second feature (AR IVT) is the most important for predicting AR precipitation, for both seasons (see code). This further justifies the assumption that we have a strong underlying AR IVT-precipitation relationship and that we can proceed by only plotting AR IVT (x-axis) versus predicted AR precipitation (y-axis). As a final step, the new scatterplots produced by the three ML models were fed into a binscatter function that calculates the conditional mean of AR precipitation based on AR IVT. This part of my code was already written for my previous research, so I only had to change the x and y data inputs to produce my final results.

Results

For the sake of not including too many figures in this report, I only show the new scatterplots for the random forest model. The scatterplots for the linear regression and SVR models can be found in my code (and time permitting, I may make a supplementary figure document). For the conditional mean plots, results for all the ML models and my existing work are included below.

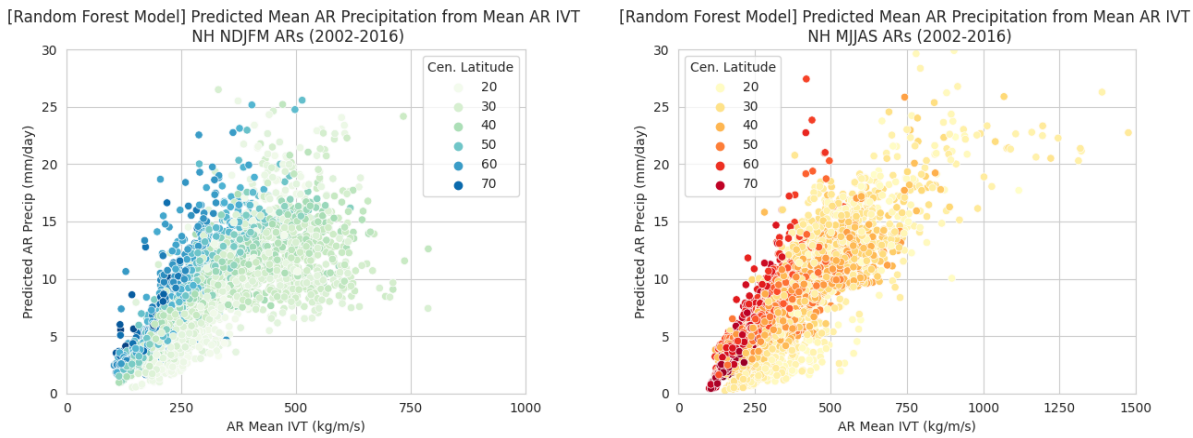

Figure 3. Mean AR IVT versus predicted mean AR precipitation, for both NH NDJFM and NH MJJAS ARs. The color hue (central latitude in degrees) is added to highlight that generally, high latitudes correspond to lower IVT values and slightly less AR precipitation.

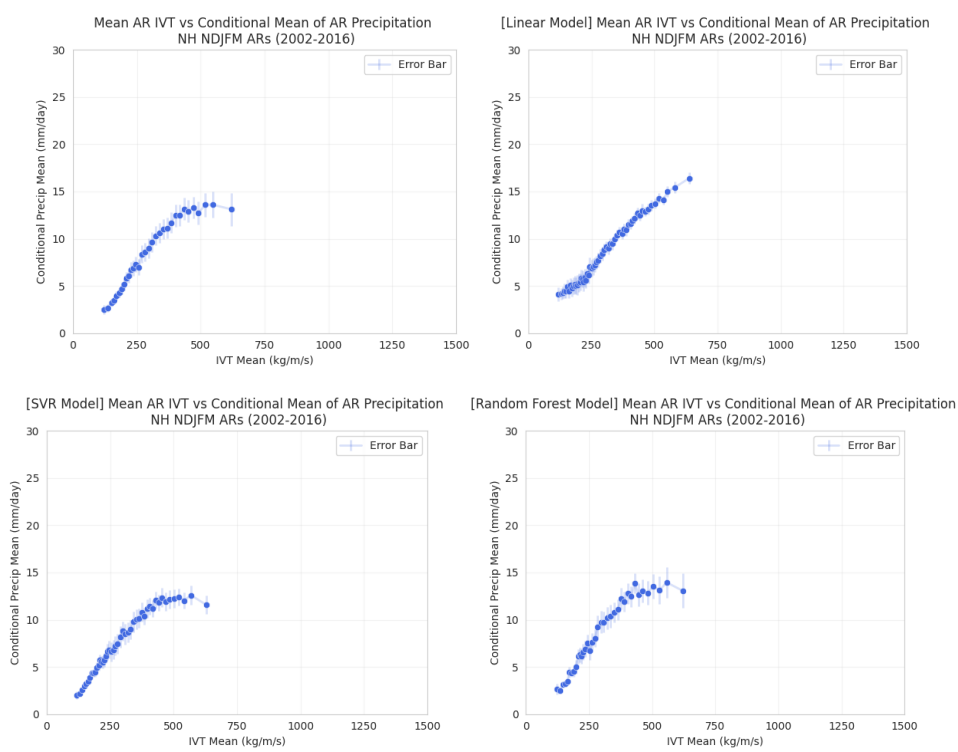

Figure 4. Mean AR IVT versus conditional mean of AR precipitation, for NDJFM ARs. Top left plot is from the original (non-ML) method, compared to linear, SVR, and random forest models.

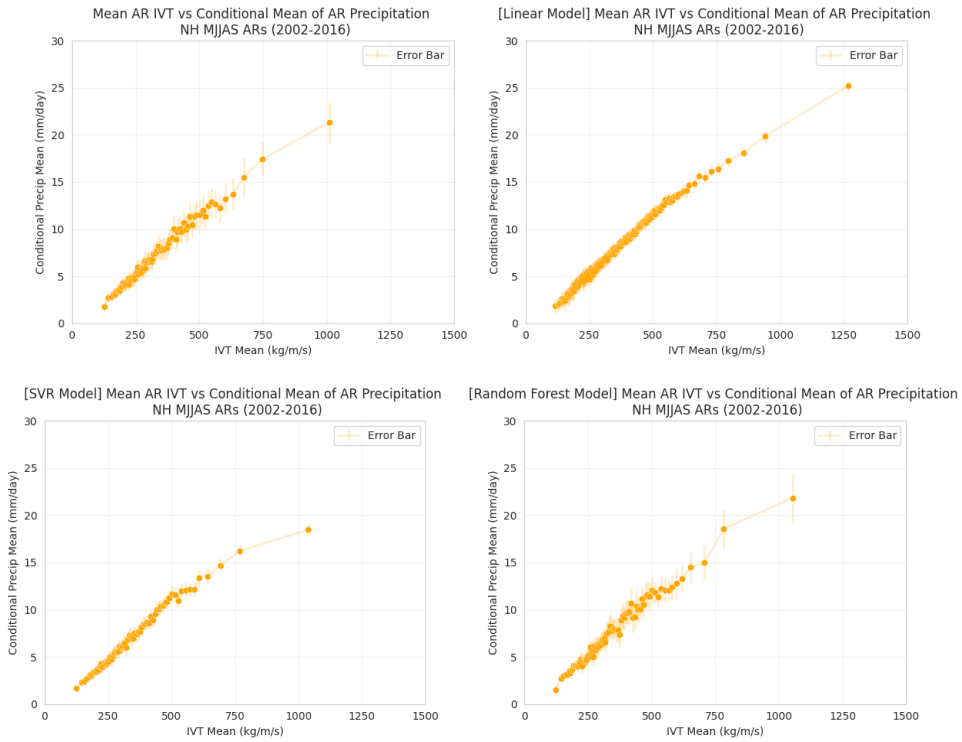

Figure 5. Mean AR IVT versus conditional mean of AR precipitation, for MJJAS ARs. Top left plot is from the original (non-ML) method, compared to linear, SVR, and random forest models.

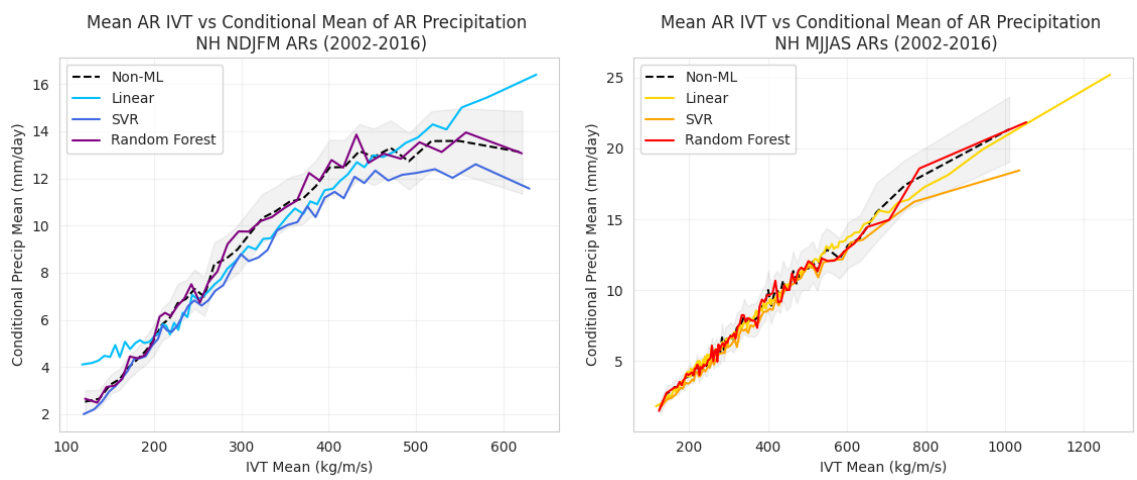

Figure 6. The same as Figures 4–5, but zoomed in and plotted simultaneously to show how the original method and ML methods compare. (Left) For NH NDJFM ARs. (Right) for NH MJJAS ARs.

Table 1. The root mean square error (RMSE)* for conditional mean AR precipitation of each ML model when compared to the original method.

Root Mean Square Error (RMSE)	**NDJFM ARs**	**MJJAS ARs**
Linear Model	3.640 mm/day	4.433 mm/day
SVR Model	2.851 mm/day	0.923 mm/day
Random Forest Model	0.894 mm/day	1.526 mm/day

**Despite setting a fixed random state for reproducible results, the exact RMSE values are slightly different each time I run my code. However, the overall patterns still stand.*

Figure 3 shows the scatter of mean AR IVT versus the predicted mean AR precipitation for the random forest model. Out of all the models, random forest regression shows the most scatter. Figures 4–6 show the results of taking the conditional mean of AR precipitation based on mean AR IVT, for my original method and the three ML algorithms. Overall, each regression model predicts a strong, AR IVT-precipitation (conditional) relationship. Lastly, Table 1 quantifies the root mean square error (RMSE) in the predicted AR precipitation of each ML algorithm when compared to the original method. Since the original (non-ML) conditional mean AR precipitation is produced from the real, raw satellite data, it can be taken as *ground truth* to define an error metric. As such, the RMSE values in Table 1 quantify the difference seen in Figure 6. Note that each conditional mean plot has error bars (corresponding to the 95% confidence interval). This comes from the uncertainty in the binscatter function since the sampling of the bin widths is dynamic (i.e., not constant sized) [1].

Discussion

The results for the random forest regression model in Figure 3 show a relationship that still has considerable scatter. However, when compared to the real data in Figure 2, one can see a clearer relationship emerge. The linear and SVR model scatterplots (not shown) have less scatter and have tight clustering, which suggests a strong, positive AR IVT precipitation relationship from the get-go. The linear model, unsurprisingly, fails to predict AR precipitation values that deviate from the overall linear pattern. However, the SVR and random forest models can predict AR precipitation values more flexibly, as evidenced by the higher scatter (e.g., Figure 3). These patterns are more apparent when looking at the conditional mean plots (Figures 4–6).

Figures 4–6 show that, remarkably, each regression model can reproduce the original conditional AR IVT-precipitation relationship quite well. For NDJFM ARs (Figure 4), the original relationship is overall linear with a small plateau for higher IVT values (>500 kg/m/s). The linear model does well in predicting this relationship but fails to pick up the plateau. The SVR and

random forest models, which inherently can deal with nonlinearity better, do pick up on the tail-end plateau and produce near-identical results. For MJJAS ARs (Figure 5), the original relationship is again overall linear, with some "instability" around the mid-IVT values (~500 kg/m/s). The linear model predicts an almost perfect linear relationship (note the small error bars too) but fails to recognize the "instability" in the mid-IVT values. The SVR and random forest models do a better job of predicting the small nonlinear deviations in the conditional relationship.

Finally, Figure 6 and Table 1 let us zoom in and quantify the differences between what the ML models predict and the *ground truth* conditional relationship between AR IVT and AR precipitation. For NDJFM ARs, the random forest model best predicts the relationship and yields a low RMSE value of <1 mm/day. The linear model has a bias for low and high IVT values, where it predicts too much precipitation for those given AR IVT values. The SVR model follows the original relationship, but for high IVT values, predicts too little precipitation (bias). These biases in the linear and SVR models contribute to higher RMSE values in Table 1. For MJJAS ARs, the random forest model still predicts the relationship well, but the SVR model does even better (by the RMSE metric). Virtually all the models predict the initial relationship correctly, and deviations in performance come from the "instabilities" (jumpy behavior) for the mid-IVT values.

Conclusions

To summarize, existing research can produce a strong, linear conditional relationship between mean AR IVT and mean AR precipitation in satellite data. Producing these conditional statistics can provide a benchmark for analyzing the AR IVT-precipitation relationships in other datasets, useful for future climate and weather studies. A machine learning approach can be used to simplify the clutter in the raw data by using supervised regression algorithms to predict AR precipitation from AR IVT. Overall, all three ML models (linear, SVR, and random forest) can reproduce the original conditional relationship with good accuracy. The linear model has a weakness in identifying nonlinear deviations whereas the SVR and random forest models can predict these deviations well. For both NDJFM and MJJAS ARs, the SVR and random forest models are superior in predicting the AR IVT-precipitation relationship and have the lowest RMSE values.

Future work should explore the conditional relationships between other AR variables (e.g., AR moisture and wind) versus AR precipitation. From my previous work, these relationships are highly nonlinear, so it would be interesting to see how the SVR and random forest models reproduce them. For further analysis, one could try using neural network regression and see how it compares to SVR and random forest. Additionally, one could optimize the parameters chosen for the models (e.g., maybe a different value for maximum tree depth would make an even better

random forest model). I would like to thank Alex for helping me through my project when my initial idea failed and Nique for providing support throughout the quarter.

References

[1] Cattaneo, M. D., Crump, R. K., Farrell, M. H., Feng, Y. (2021). On Binscatter. *arXiv Economics*. https://arxiv.org/abs/1902.09608.

[2] Gonzales, K. R., Swain, D. L., Barnes, E. A., & Diffenbaugh, N. S. (2020). Moisture- versus wind-dominated flavors of atmospheric rivers. Geophysical Research Letters, 47, e2020GL090042. https://doi.org/10.1029/2020GL090042

[3] Peters, O., Neelin, J. (2006). Critical phenomena in atmospheric precipitation. *Nature Phys* 2, 393–396.